

Day 5 —SPL Cheat Sheet — Splunk Free

Name: Mohammed Farhan

Role: Associate – Information Security (Intern)

Organization: CyberShelter UAE

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Environment

Source	Sourcetype	Method
Linux auth.log	linux_auth	Continuous monitoring (live)
Windows Security + Sysmon	WinEventLog:Security / WinEventLog:Microsoft-Windows-Sysmon/Operational	Universal Forwarder (live, real-time)
Kali firewall log	ufw_firewall	Static one-time snapshot

Part 1 — Splunk Installation on Ubuntu Server

Downloaded Splunk :

- `wget-Osplunk.deb "https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linuxamd64.deb"`

Installed:

- `sudo dpkg -i splunk.deb`

```

fara@ubuntu:~$ wget -O splunk.deb "https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.deb"
--2026-07-03 13:53:46-- https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com) ... 108.139.86.27, 108.139.86.34, 108.139.86.117, ...
Connecting to download.splunk.com (download.splunk.com)|108.139.86.27|:443 ...
connected.
HTTP request sent, awaiting response... 200 OK
Length: 1322956776 (1.2G) [binary/octet-stream]
Saving to: 'splunk.deb'

splunk.deb      100%[=====>]  1.23G  720KB/s   in 45m 10s

2026-07-03 14:38:58 (477 KB/s) - 'splunk.deb' saved [1322956776/1322956776]

fara@ubuntu:~$ ls -lh splunk.deb
-rw-rw-r-- 1 fara fara 1.3G Jun 26 01:08 splunk.deb
fara@ubuntu:~$

```

Started Splunk (needed `--run-as-root` since running as root is blocked by default):

- `sudo /opt/splunk/bin/splunk start --accept-license --run-as-root`

```

fara@ubuntu:~$ sudo /opt/splunk/bin/splunk start --accept-license --run-as-root

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: farhan
Password must contain at least:
    * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Copying '/opt/splunk/etc/openldap/ldap.conf.default' to '/opt/splunk/etc/openldap/ldap.conf'.
writing RSA key

writing RSA key

Moving '/opt/splunk/share/splunk/search_mrsparkle/modules.new' to '/opt/splunk/share/splunk/search_mrsparkle/modules'.

```

Verified running:

- `sudo /opt/splunk/bin/splunk status`

Confirmed splunkd running, web interface on port 8000.

- `sudo ss -tulnp | grep splunk`

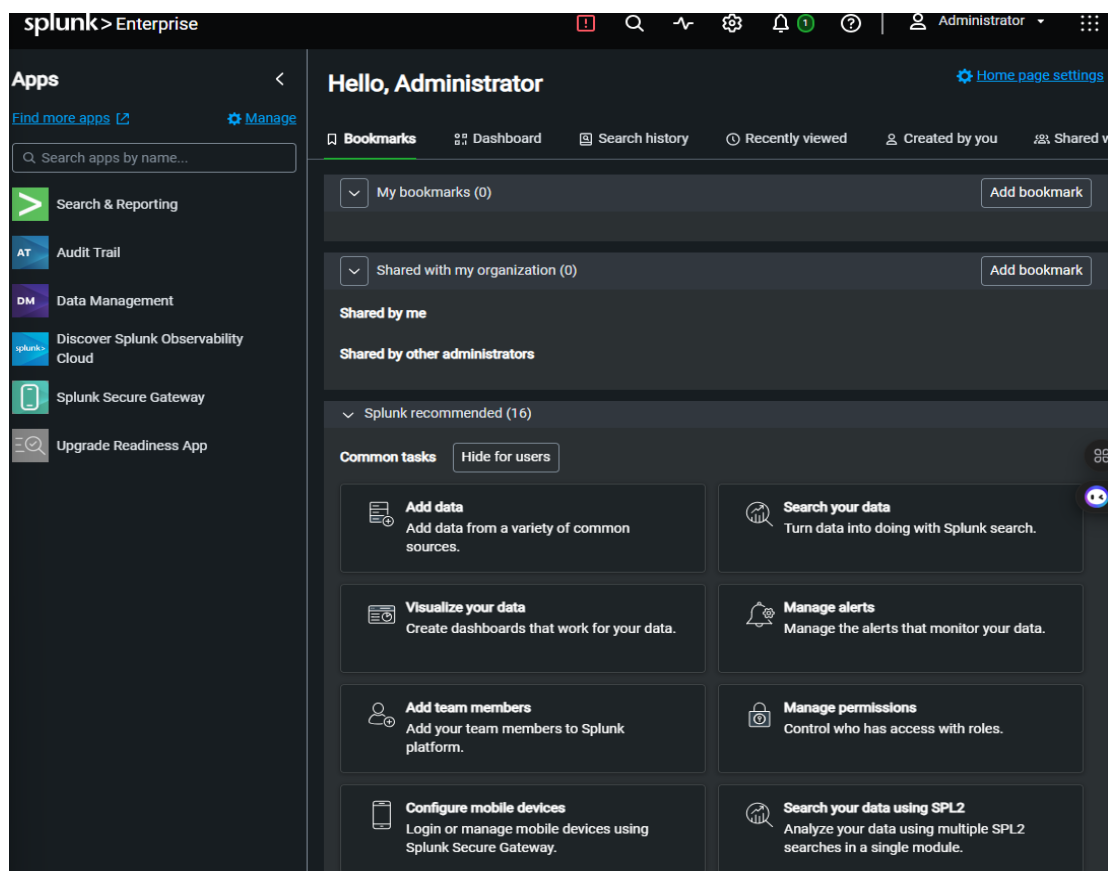
Opened firewall port:

- `sudo ufw allow 8000/tcp`

```
fara@ubuntu-server:~$ sudo ufw allow 8000/tcp
Rule added
Rule added (v6)
fara@ubuntu-server:~$
```

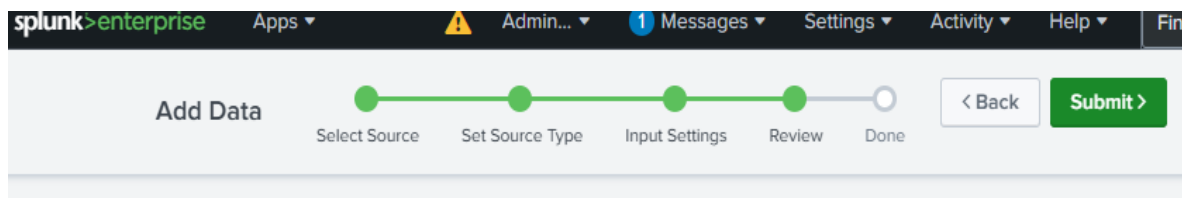
Accessed and logged in:

- <http://192.168.56.104:8000>



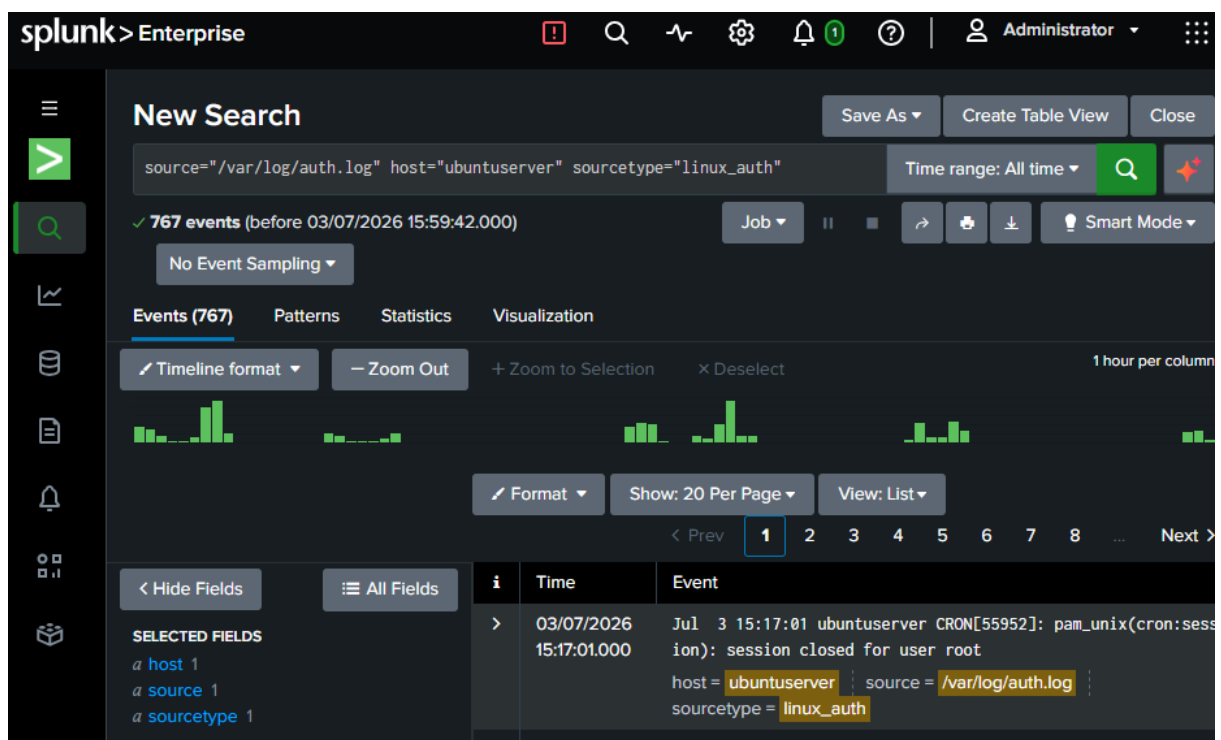
Part 2 — Log Source Ingestion (Initial Static Approach)

Source 1 — Linux auth.log: Monitored directly on Ubuntu Server via Splunk's "Add Data → Monitor → Files & Directories," continuously monitoring `/var/log/auth.log`, sourcetype `linux_auth`.



Review

Input Type File Monitor
Source Path /var/log/auth.log
Continuously Monitor Yes
Source Type linux_auth
App Context search
Host ubuntuserver
Index default



Source 2 — Windows Security XML: Generated test events on Windows 10 (failed login, PowerShell execution, new user creation, group membership change), exported Security log via Event Viewer as windows-sec-logs.xml, transferred via SCP to Ubuntu Server, ingested as one-time index with sourcetype windows_security_xml (log4net_xml parsing, host=windows10).

splunk>enterprise

Apps

Administrator

1 Messages

Settings

Activity

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Submit >

Review

Input Type

Source Path

Continuously Monitor

Source Type

App Context

Host

Index

File Monitor

/home/fara/windows-sec-logs.xml

No, index once

windows_security_xml

search

windows10

default

New Search

Save As

Create Table

source="/home/fara/windows-sec-logs.xml" host="windows10" sourcetype="windows_security_xml"

Time range: Today

337,275 events (03/07/2026 00:00:00.000 to 03/07/2026 17:38:25.000)

No Event Sampling

Job

Events (337,275)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

+ Zoom to Selection

x Deselect

Format

Show: 20 Per Page

View: List

< Prev

1

2

3

4

5

6

7

< Hide Fields

All Fields

	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1	>	03/07/2026 17:21:56.000	Security ID: S-1-5-18 host = windows10 source = /home/fara/windows-sec-logs.xml sourcetype = windows_security_xml
	>	03/07/2026 17:21:56.000	Creator Subject: host = windows10 source = /home/fara/windows-sec-logs.xml sourcetype = windows_security_xml

Source 3 — Firewall log: After multiple failed attempts to find a working public sample dataset, used Kali's own real UFW log (/var/log/ufw.log) — copied to Desktop, transferred via SCP to Ubuntu Server, ingested as one-time index, sourcetype ufw_firewall, host=kali.

Review

```
Input Type ..... File Monitor
Source Path ..... /home/fara/sample-firewall.log
Continuously Monitor ..... No, index once
Source Type ..... ufw_firewall
App Context ..... search
Host ..... kali
Index ..... default
```

		host = kali	source = /home/fara/sample-firewall.log	sourcetype = ufw_firewall
>	01/07/2026 00:35:40.181	2026-07-01T02:05:40.181574+05:30 kali kernel: [UFW BLOCK] IN=eth1 OUT= MAC= SRC=fe80:0000:0000:0000:0a00:27ff:fe5f:c3ef DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=735467 PROTO=UDP SPT=34196 DPT=3702 LEN=615		
		host = kali	source = /home/fara/sample-firewall.log	sourcetype = ufw_firewall
>	01/07/2026 00:35:40.177	2026-07-01T02:05:40.177086+05:30 kali kernel: [UFW BLOCK] IN=eth1 OUT= MAC= SRC=192.168.56.102 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=15431 DF PROTO=UDP SPT=45060 DPT=3702 LEN=615		
		host = kali	source = /home/fara/sample-firewall.log	sourcetype = ufw_firewall
>	30/06/2026 17:50:55.055	2026-06-30T19:20:55.055253+05:30 kali kernel: [UFW BLOCK] IN=eth0 OUT= MAC=08:00:27:47:b8:1c:52:55:0a:00:02:02:08:00 SRC=54.39.128.230 DST=10.0.2.15 LEN=40 TOS=0x08 PREC=0x00 TTL=64 ID=55909 PROTO=TCP SPT=80 DPT=57738 WINDOW=65535 RES=0x00 ACK URG=0		
		host = kali	source = /home/fara/sample-firewall.log	sourcetype = ufw_firewall

SPL Cheat Sheet

Search 1 — Login Activity (Failed & Successful)

```
spl
index=main sourcetype=linux_auth "Failed password"
```

index=main "failed password" Time range: Last 15 minutes

✓ 1 event (03/07/2026 18:31:27.000 to 03/07/2026 18:46:27.000) Job

No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Format Show: 20 Per Page View: List

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS		>	03/07/2026 18:46:02.000	Jul 3 18:46:02 ubuntuuser sshd[77422]: Failed password for fara from 192.168.56.102 port 34682 ssh2 host = ubuntuuser source = /var/log/auth.log sourcetype = linux_auth

host 1
source 1
sourcetype 1

Detects: Brute force login attempts

MITRE ATT&CK: T1110 — Brute Force

spl

index=main sourcetype=linux_auth "accepted password"

index=main accepted password Time range: Last 60 minutes

✓ 1 event (03/07/2026 17:52:00.000 to 03/07/2026 18:52:09.000) Job

No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Format Show: 20 Per Page View: List

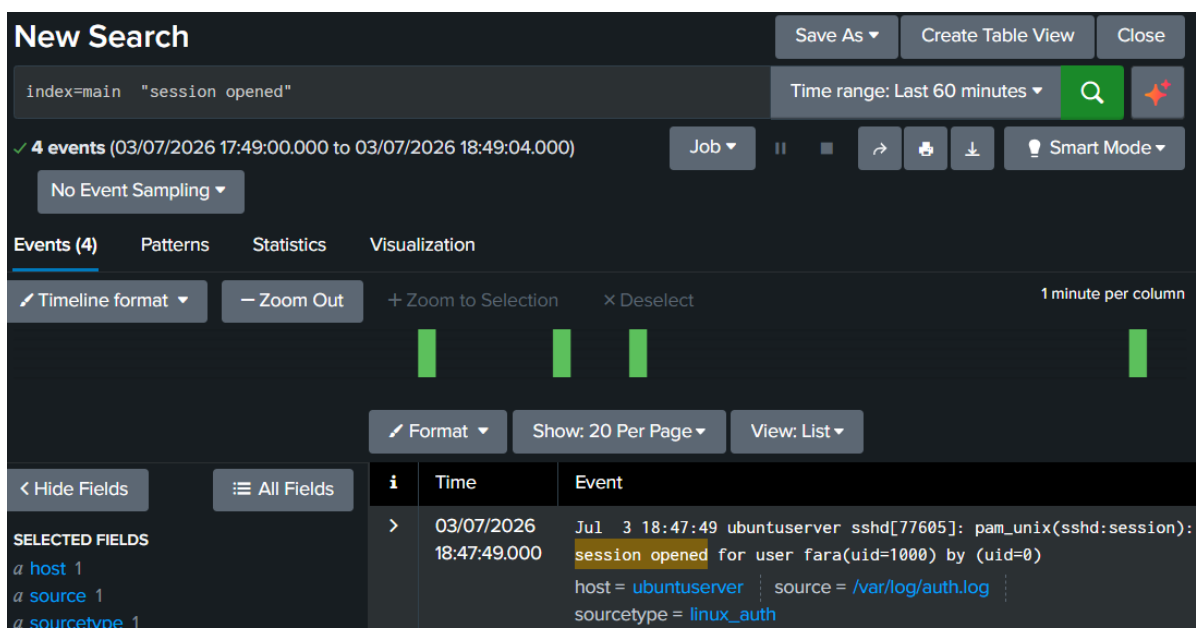
< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS		>	03/07/2026 18:47:49.000	Jul 3 18:47:49 ubuntuuser sshd[77605]: Accepted password for fara from 192.168.56.102 port 41694 ssh2 host = ubuntuuser source = /var/log/auth.log sourcetype = linux_auth

host 1
source 1
sourcetype 1

Detects: Successful SSH logins

MITRE ATT&CK: T1078 — Valid Accounts

index=main sourcetype=linux_auth "session opened"



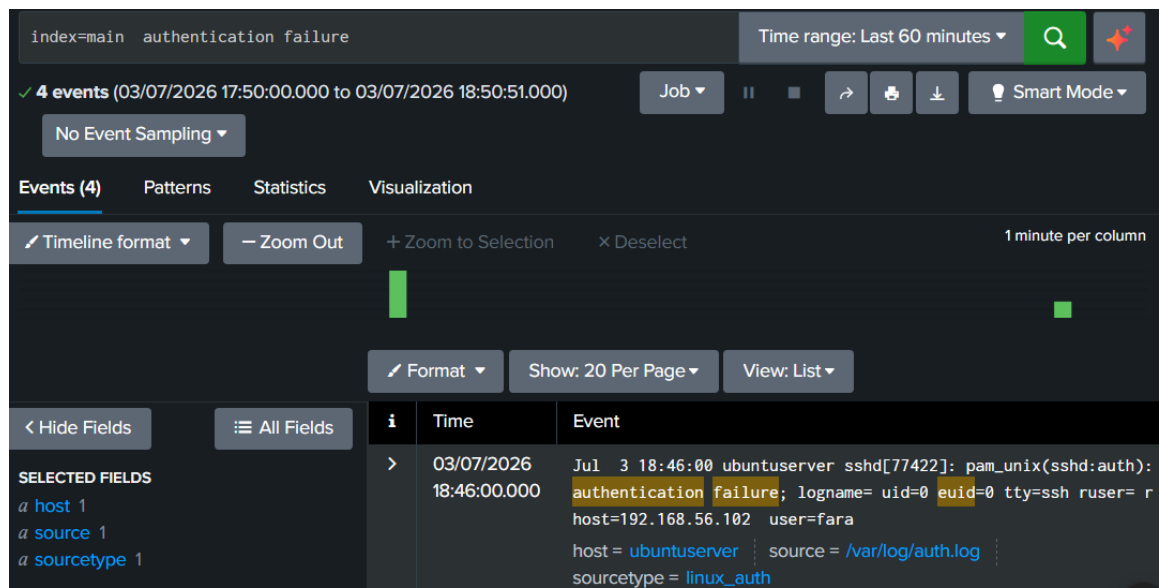
+

Detects: Baseline of all successful sessions — used to compare against failed attempts and spot anomalies

MITRE ATT&CK: T1078 – Valid Accounts

spl

index=main sourcetype=linux_auth "authentication failure"



Detects: Broader PAM-level authentication failures

MITRE ATT&CK: T1110 — Brute Force

Search 2 — New User Account Creation

spl

index=main "EventID>4720" testuser 1

New Search

index=main "EventID>4720" testuser1

Time range: Last 4 hours

1 event (03/07/2026 15:17:00.000 to 03/07/2026 19:17:09.000)

No Event Sampling

Job

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 minute per column

Format Show: 20 Per Page View: List

	i	Time	Event
	>	03/07/2026 17:21:53.000	Account Domain: FARHANSOC</Message><Level>Information</Level><Task>User Account Management</Task><Opcode>Info</Opcode><Channel>Security</Channel><Provider>Microsoft Windows security auditing.</Provider><Keywords><Keyword>Audit Success</Keyword></Keywords></RenderingInfo></Event><Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4720</EventID><Version>0</Version><Level>0</Level><Task>13824</Task><Opcode>0</Opcode><Keywords>0x802000000000</Keywords><TimeCreated SystemTime='2026-07-03T10:06:44.8433954Z' /><EventRecordID>20106</EventRecordID><Correlation ActivityID='{122fdd2e-0b2f-0000-ecdd-2f122f0bdd01}' /><Execution ProcessID='676' ThreadID='3516' /><Channel>Security</Channel><Computer>farhansoc.soc.lab</Computer><Security></System><EventData><Data Name='TargetUserName'>testuser1</Data><Data Name='TargetDomainName'>FARHANSOC</Data><Data Name='TargetSid'>S-1-5-21-3821258635-4097038788-1053219664-1002</Data><Data Name='SubjectUserSid'>S-1-5-21-3821258635-4097038788-1053219664-1001</Data><Data Name='SubjectUserName'>farhan</Data><Data Name='SubjectDomainName'>FARHANSOC</Data><Data Name='SubjectLogonId'>0xd2b4f</Data><Data Name='PrivilegeList'></Data><Data Name='SamAccountName'>testuser1</Data><Data Name='DisplayName'>%*1793</Data><Data Name='UserPrincipalName'></Data><Data Name='HomeDirectory'>%*1793</Data><Data Name='HomePath'>%*1793</Data><Data Name='ScriptPath'>%*1793</Data><Data Name='ProfilePath'>%*1793</Data><Data Name='UserWorkstations'>%*1793</Data><Data Name='PasswordLastSet'>%*1794</Data><Data Name='AccountExpires'>%*1794</Data><Data Name='PrimaryGroupId'>513</Data><Data Name='AllowedToDelegateTo'></Data><Data Name='OldUacValue'>0x0</Data><Data Name='NewUacValue'>0x15</Data><Data Name='UserAccountControl'></Data></EventData></Event>

host = windows10 | source = /home/fara/windows-sec-logs.xml | sourcetype = windows_security_xml

Detects: New user account created — attackers create backdoor accounts to maintain persistent access

MITRE ATT&CK: T1136 — Create Account

Search 4 — DNS Queries

spl

index=main EventCode=22 | top QueryName

The screenshot shows the Splunk Search interface for a search named "New Search". The search query is `index=* EventCode=22 | top QueryName`. The time range is set to "Last 60 minutes". The search results show 34 events from 04/07/2026 00:57:00.000 to 04/07/2026 01:57:33.000. The "Statistics (10)" tab is selected, displaying a table of DNS query names ranked by frequency.

QueryName	count	percent
wpad	10	29.411
farhansoc	6	17.647
FARHANSOC	3	8.823
www.msftconnecttest.com	1	2.941
onedcolprdcus05.centralus.cloudapp.azure.com	1	2.941
microsoft.com	1	2.941
keagegmmhdhhee	1	2.941
ipv6.msftconnecttest.com	1	2.941
img-s-msn-com.akamaized.net	1	2.941

Detects: DNS queries made by processes, ranked by frequency — low-frequency/rare domains can indicate C2 (command and control) communication

MITRE ATT&CK: T1071.004 — Application Layer Protocol: DNS

Search 6 — PowerShell Executions

spl

```
index=main sourcetype=WinEventLog:Security "powershell.exe"
```

The screenshot shows the Splunk Search interface. At the top, the search bar contains the query `index=main sourcetype=WinEventLog:Security "powershell.exe"`. The time range is set to "Last 60 minutes". Below the search bar, it indicates "4 events" found. The interface includes tabs for "Events (4)", "Patterns", "Statistics", and "Visualization". The "Events" tab is active, showing a list of events. The first event is expanded, showing details for a PowerShell execution on 07/04/2026 at 05:55:54 AM. The event details include the LogName=Security, Process ID: 0x2638, and Process Name: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe. The interface also features a sidebar with "SELECTED FIELDS" (host, source, sourcetype) and "INTERESTING FIELDS" (Account_Domain, Account_Name).

New Search

Save As Create Table View Close

index=main sourcetype=WinEventLog:Security "powershell.exe"

Time range: Last 60 minutes

4 events (04/07/2026 05:11:00.000 to 04/07/2026 06:11:40.000)

No Event Sampling

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 minute per column

Format Show: 20 Per Page View: Raw

< Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 1
- sourcetype 1

INTERESTING FIELDS

- Account_Domain 1
- Account_Name 1

Event

> 07/04/2026 05:55:54 AM

LogName=Security

... 22 lines omitted ...

Process Information:

- Process ID: 0x2638
- Process Name: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

Show all 27 lines

Detects: PowerShell execution — commonly abused for "living-off-the-land" attacks since it's a trusted, legitimate Windows tool

MITRE ATT&CK: T1059.001 — Command and Scripting Interpreter: PowerShell

Search 7 — Account Lockouts

Configuration: PAM faillock configured on Ubuntu Server (`/etc/pam.d/common-auth`, `/etc/pam.d/common-account`, `/etc/security/faillock.conf`) to lock accounts after repeated failed SSH attempts.

spl

index=main "account locked"

New Search

Save As>Create Table View>Close

index=main account locked

Time range: Last 4 hours>

Q

✓ 1 event (03/07/2026 17:58:00.000 to 03/07/2026 21:58:31.000)

Job>||

→

🖨

⬇

Verbose Mode>

No Event Sampling>

Events (1)PatternsStatisticsVisualization

Timeline format>

Zoom Out

+ Zoom to Selection

× Deselect

1 minute per column

Format>

Show: 20 Per Page>

View: List>

< Hide Fields

All Fields

i

Time

Event

>

03/07/2026 20:46:14.000

Jul 3 20:46:14 ubuntuuser sshd[93180]: pam_faillock (sshd:auth): Consecutive login failures for user fara account temporarily locked
host = ubuntuuser source = /var/log/auth.log
sourcetype = linux_auth

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

Detects: Account lockout triggered by repeated failed login attempts

MITRE ATT&CK: T1110 — Brute Force

Search 8 — Top Source IPs

spl
index=main | top src_ip

New Search

index=main | top src_ip

✓ 189 events (03/07/2026 18:16:00.000 to 03/07/2026 22:16:22.000)

No Event Sampling>

Job

Events (189)PatternsStatistics (2)Visualization

Show: 20 Per Page>

Format>

Preview: On

src_ip

count

192.168.56.10218

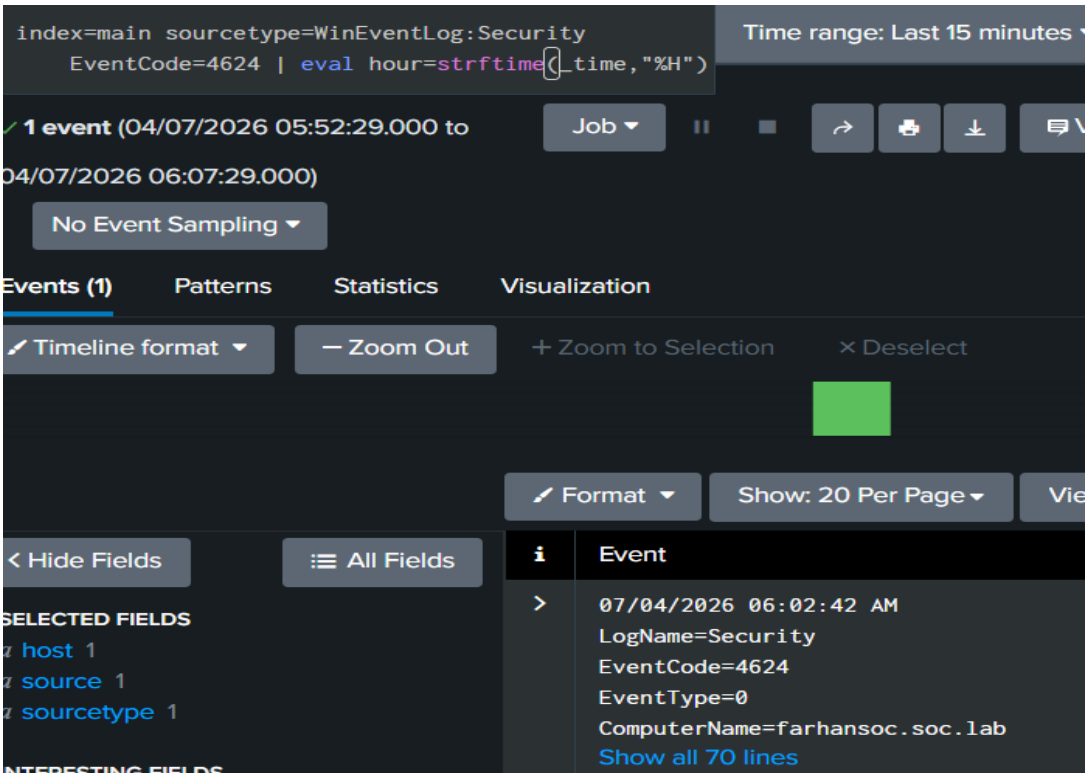
fara2

Detects: Most frequent source IPs connecting via SSH — identifies potential scanning or repeated brute force sources

MITRE ATT&CK: T1595 — Active Scanning

Search 9 — Logon Success Outside Business Hours

index=main sourcetype=WinEventLog:Security EventCode=4624 | eval
hour=strftime(_time, "%H")



Detects: Successful logons occurring outside normal business hours (before 9 AM or after 6 PM) — indicator of compromised credentials being used at unusual times

MITRE ATT&CK: T1078 — Valid Accounts

Search 10 — Changes to Privileged Groups

spl

index=main sourcetype=WinEventLog:Security EventCode=4732

The screenshot shows the Splunk search interface with the query `index=main sourcetype=WinEventLog:Security EventCode=4732`. The search results show 2 events from 04/07/2026 05:42:27.000 to 04/07/2026 05:57:27.000. The interface includes tabs for Events (2), Patterns, Statistics, and Visualization. The Events tab is active, showing a timeline format. The search results are displayed in a table with columns for Event, LogName, EventCode, EventType, ComputerName, SourceName, Type, RecordNumber, Keywords, TaskCategory, OpCode, and Message. The message indicates that a member was added to a security-enabled local group.

Event
> 07/04/2026 05:55:54 AM LogName=Security EventCode=4732 EventType=0 ComputerName=farhansoc.soc.lab SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=21463 Keywords=Audit Success TaskCategory=Security Group Management OpCode=Info Message=A member was added to a security-enabled local group. Subject: Security ID: S-1-5-21-3821258635-4097038788-1053219664-1001 Account Name: farhan Account Domain: FARHANSOC Logon ID: 0x74700

Detects: User added to a privileged/security-enabled group — classic privilege escalation technique

MITRE ATT&CK: T1098 — Account Manipulation

Conclusion

Eight of ten planned SPL searches were successfully built, tested, and confirmed with live data across Linux authentication logs, Windows Security events, and Sysmon telemetry. Real-world troubleshooting was required throughout the day. These fixes transformed static log snapshots into genuine live data pipelines, closely mirroring how log ingestion issues are diagnosed and resolved in real production SOC environments.